

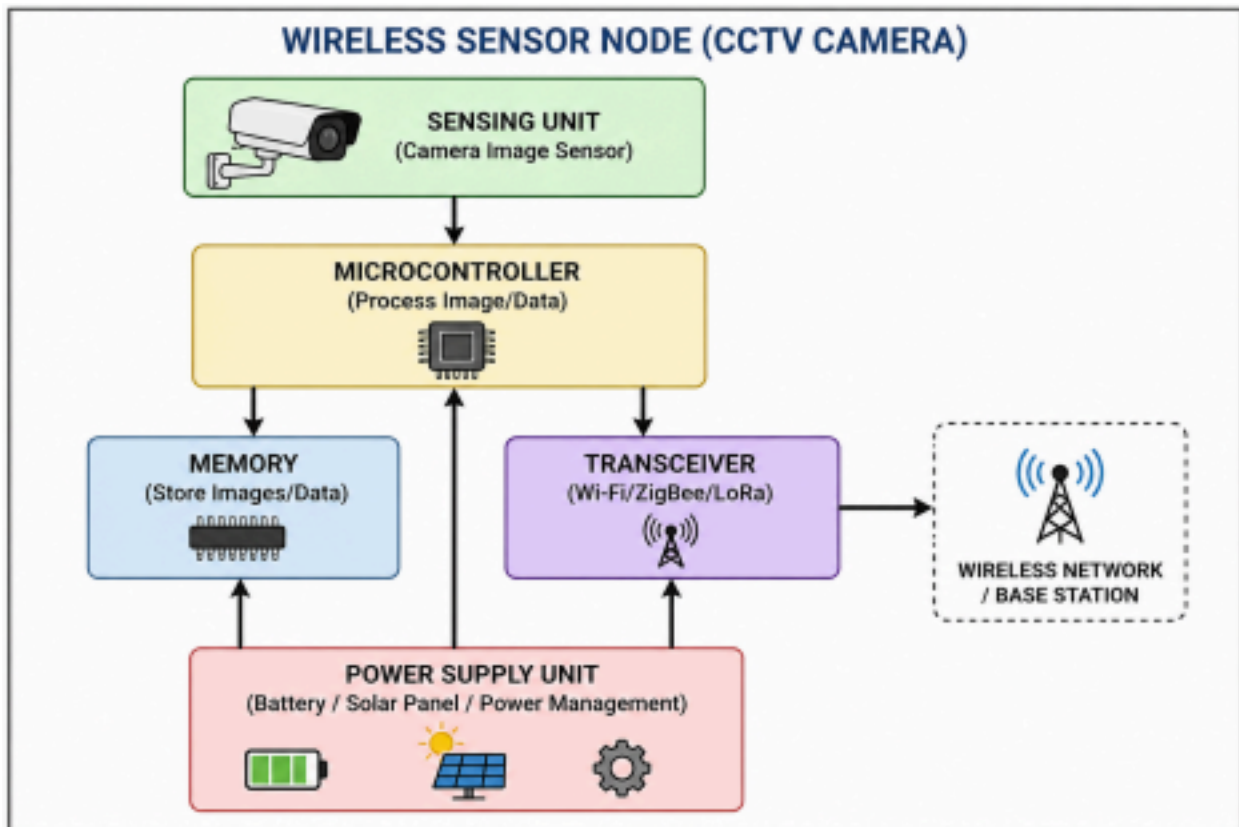
Practical 1 - Study of Camera Sensor Node Hardware Architecture

SUB WIRELESS SENSOR NETWORK

AIM:WORKING ON CAMERA SENSOR

Design a conceptual block diagram of a wireless sensor node and analyse the role of sensing unit, microcontroller, transceiver, memory, and power supply in energy-constrained environments.

BLOCK DIAGRAM FOR CAMERA SENSOR



Camera-Based Wireless Sensor Node

Example: CCTV Camera

- **Sensing Unit:** Captures images and videos.
- **Microcontroller:** Processes the captured data.
- **Memory:** Stores images/videos temporarily.
- **Transceiver:** Sends data wirelessly to the monitoring station.
- **Power Supply:** Provides energy using battery or solar power.

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Role of Each Component in Energy

1. Sensing Unit

- Captures images or videos from the environment.
- Converts physical information into digital signals.
- Consumes energy while continuously sensing.
- To save energy, motion detection or periodic sensing is used.

Role in CCTV: Captures video only when motion is detected to reduce power consumption.

2. Microcontroller

- Acts as the brain of the sensor node.
- Controls all hardware components.
- Processes sensor data before transmission.
- Uses sleep mode when no processing is required to save battery. Role in

CCTV: Compresses image/video and detects motion before sending data.

3. Memory

- Stores sensor readings, images, or temporary data.
- Flash memory is used for permanent storage.
- RAM is used during processing.
- Efficient memory management reduces energy consumption.

Role in CCTV: Stores captured frames before transmitting them.

4. Transceiver

- Sends and receives data wirelessly.
- Uses technologies like Wi-Fi, ZigBee, Bluetooth, or LoRa.
- Communication consumes the highest amount of energy.
- Energy is saved by transmitting only important data.

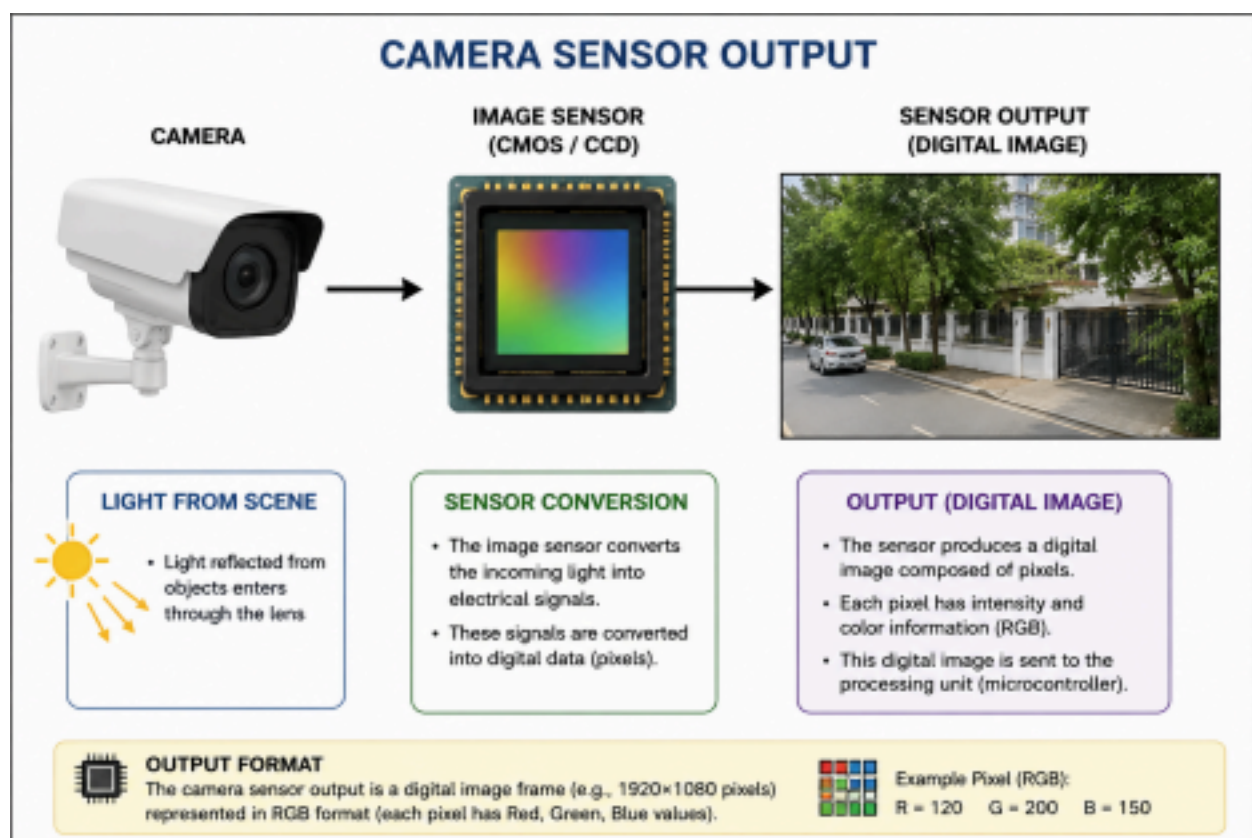
Role in CCTV: Sends motion-detected video instead of continuous streaming.

5. Power Supply

- Supplies energy to the entire sensor node.
- Can use batteries, solar panels, or energy harvesting.
- Power management techniques increase battery life.
- Sleep mode helps reduce energy consumption.

Role in CCTV: Solar-powered CCTV cameras can work for long periods without frequent battery replacement

OUTPUT:





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